Lesson 1: The Cartesian Plane, Growing Patterns, and Types of Variables

Plotting and naming points, describing a growing pattern with a table, a graph and an equation, and telling discrete variables from continuous ones.

1. The Cartesian Coordinate System

The word "Cartesian" honours the French mathematician and philosopher **René Descartes** (1596–1650), who in 1637 introduced the idea of locating a point on a flat surface using two perpendicular reference lines.

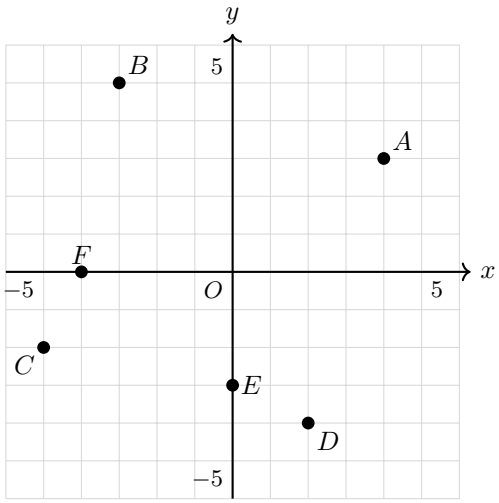
A **Cartesian coordinate system** consists of two number lines, called **axes**, that cross at right angles. The horizontal axis is the x-**axis** and the vertical axis is the y-**axis**; together they form the xy-plane, also called the **Cartesian plane**. The point where the two axes cross is called the **origin**, usually labelled O.

Every point on the plane can be located by a unique **ordered pair** of numbers, called its **coordinates**. The first number, the ***x*-coordinate**, tells how far to move horizontally from the origin; the second number, the ***y*-coordinate**, tells how far to move vertically. For instance, the point $P(5, 2)$ is located by moving 5 units right and 2 units up from the origin.

The two axes divide the plane into four regions called **quadrants**, numbered I through IV starting at the top-right region and moving counter-clockwise.

Example 1 — Reading coordinates and identifying quadrants

Refer to the grid below.



(a) Complete the following by writing the coordinates of each labelled point.

A(_____) **B**(_____)
C(_____) **D**(_____)
E(_____) **F**(_____)

(b) State the coordinates of the point in Quadrant II.

(c) State the coordinates of the point in Quadrant IV.

(d) Complete the table below using "positive" or "negative".

Quadrant	x**-coordinate**	y**-coordinate**
I		
II		
III		
IV		

2. Describing a Growing Pattern With a Relation

Many real situations produce a pattern of numbers that grows by the same amount at every step. A pattern like this can always be described in three equivalent ways: a **table of values**, a **graph**, and an **equation**.

A single square is built using 4 matchsticks, one for each side. Each additional square is attached along one full side of the previous square, so it needs only 3 new matchsticks to complete its remaining three sides. The pattern continues in this way, forming a row of adjoining squares.



Figure 1 Figure 2 Figure 3

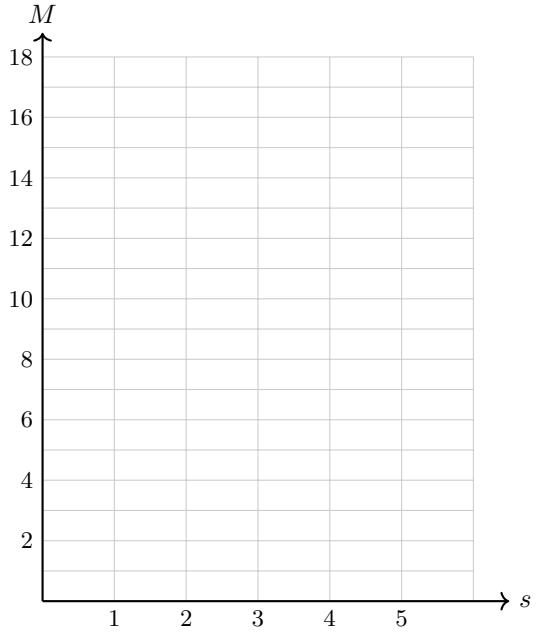
Example 2 — The matchstick pattern: diagrams, table, graph

(a) Draw the next two diagrams (Figure 4 and Figure 5) in the pattern.

(b) Complete the table relating the number of matchsticks, M, to the number of squares, s.

Number of Squares, s	1	2	3	4	5
Number of Matchsticks, M	4				

(c) Represent the data from the table of values on the grid.



Example 3 — The same pattern: why not to join, and the equation

Continue working with the matchstick pattern from Example 2.

(d) Explain why it does not make sense to join the plotted points with a straight line.

(e) Describe in words the relationship between the number of matchsticks and the number of squares.

(f) Write an equation that can be used to determine the number of matchsticks, M, if we know the number of squares, s.

(g) Use the equation to determine the number of matchsticks needed for 34 squares.

(h) Use the equation to determine the number of squares if there are 91 matchsticks.

3. Continuous Data and Linear Relations

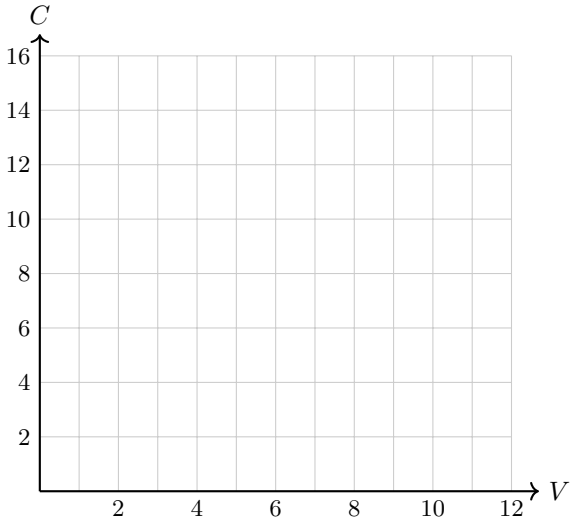
Not every relation between two variables behaves like the matchstick pattern above. Some relations involve variables that can take on *any* value in between two given values — such relations produce a graph made of connected points rather than separate dots.

Example 4 — Cost of gasoline

A gas station sells gasoline for a fixed price per litre. The table below shows the cost, C dollars, of purchasing V litres of gasoline.

Volume, V **(L)**	2	4	6	8	10
Cost, C **(\$)**	2.90	5.80	8.70	11.60	14.50

(a) Represent the data from the table on the grid.



(b) Does it make sense to join the points together with a straight line? Explain.

(c) Use the graph to estimate the cost of purchasing 5 L of gasoline.

(d) Use the graph to estimate the volume of gasoline purchased if the cost was \$10.15.

(e) Write an equation expressing C in terms of V.

(f) Use the equation to check the accuracy of your estimates in (c) and (d).

(g) In the context of the question, what does the value 1.45 represent in your equation?

4. Discrete and Continuous Variables

In Example 4, the volume of gasoline purchased and its cost are examples of **continuous variables**: it is possible to purchase, say, 2.37 L of gasoline, and it is always possible to find a value in between any two given amounts.

By contrast, in the matchstick pattern of Examples 2 and 3, the number of squares and the number of matchsticks can only take on whole-number values — there is no such thing as "square number 2.5." Variables like these, which can take only **separate, countable values with gaps between them**, are called **discrete variables**. (A discrete variable need not have a largest value — a book can have any whole number of pages — but between two allowed values there are gaps.)

When two continuous variables are graphed, it makes sense to connect the plotted points with a line or curve, since every in-between value has meaning. When two discrete variables are graphed, the points are left unconnected, since the in-between values do not correspond to anything meaningful in the situation.

Example 5 — Classifying variables

Classify each of the following variables as discrete or continuous.

(a) number of pages in a book

(b) time taken to run a marathon

(c) number of siblings a person has

(d) volume of water in a bathtub

Lesson 2: Describing Relationships Between Two Quantities

Six ways to describe a relation — words, table of values, ordered pairs, arrow diagram, equation, graph — and the independent/dependent pair behind all six.

1. Relations and Their Six Representations

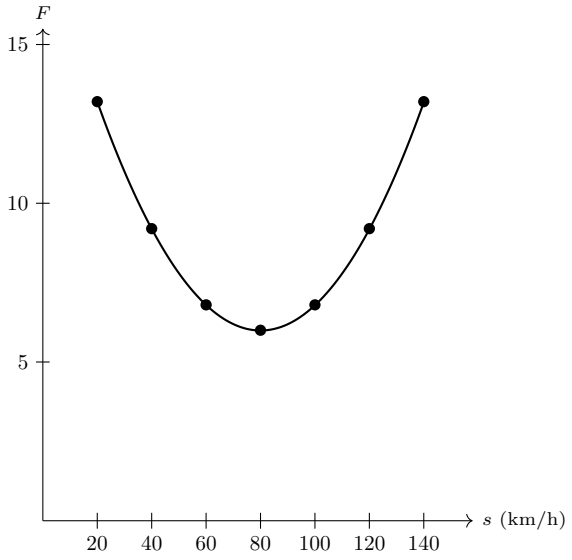
A comparison between two sets of elements is called a **relation**: the fuel remaining in a lawn mower's tank is related to how long it has been running, the monthly cost of a streaming service to the number of screens it allows, the resale value of a bicycle to its age. Throughout this unit a relation is described in six ways.

- in words
- a table of values
- a set of ordered pairs
- an arrow diagram (mapping diagram)
- an equation
- a graph

2. Investigating a Relation

Consider the relation **"the fuel consumption rate, F (litres per 100 km), of a delivery van is related to its speed, s (km/h)."** Fuel consumption *depends on* speed, so F is the **dependent variable** and s is the **independent variable**. Values of the independent variable are the **inputs**; values of the dependent variable are the **outputs**. Test data collected at $20, 40, 60, \ldots, 140$ km/h gives the table and graph below.

Input (s)	Output (F)
20	13.2
40	9.2
60	6.8
80	6.0
100	6.8
120	9.2
140	13.2



Ordered pairs (s, F): $(20, 13.2)$, $(40, 9.2)$, $(60, 6.8)$, $(80, 6.0)$, $(100, 6.8)$, $(120, 9.2)$, $(140, 13.2)$.
An output exists for *every* speed between the smallest and largest tested, not just the seven chosen, so the plotted points are joined with a smooth curve. An equation that closely fits the data is

$$F = 0.002s^2 - 0.32s + 18.8.$$

The equation is only meaningful for the speeds that make up the **domain** of the relation — it would not make sense to use it for, say, $s = -40$. The set of all output values is the **range** of the relation.

3. Locating the Independent and Dependent Variable

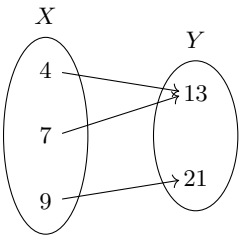
- In an **ordered pair**, the first coordinate is a value of the independent variable and the second coordinate is a value of the dependent variable.
- In a **table of values**, the independent variable is listed first — to the left of, or above, the dependent variable.
- In an **arrow diagram** (mapping diagram), arrows point from the independent variable to the dependent variable.
- On a **graph**, the independent variable is on the horizontal axis and the dependent variable is on the vertical axis.
- In an **equation**: when the equation is solved for one variable, that variable is usually the dependent one — but the context decides. (An equation such as $4x - 3y + 24 = 0$ is not solved for either variable.)

$$\overbrace{y}^{\substack{\text{dependent variable}\\ \text{(the output)}}} = \underbrace{-2x + 9}_{\substack{\text{independent variable } x \text{ sits}\\ \text{inside this expression (the input)}}}$$

Example 1 — Variables and lists from a diagram, a table and a set of pairs

For each relation, state **(i)** the independent variable, **(ii)** the dependent variable, **(iii)** the list of inputs, and **(iv)** the list of outputs.

(a)



(b)

n	d
5	18
12	24
30	9

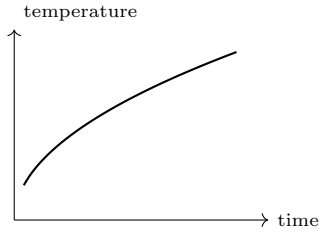
(c) (D, m):
$(2, 9),\ (5, 21),\ (8, 33),\ (11, 45)$

Example 2 — Variables from an equation, a graph and a sentence

For each relation, state **(i)** the independent variable and **(ii)** the dependent variable.

(a) $d = 4.9t^2$

(b)



(c) The number of visitors, v, to a museum on a given day is related to the price, p, of admission that day.

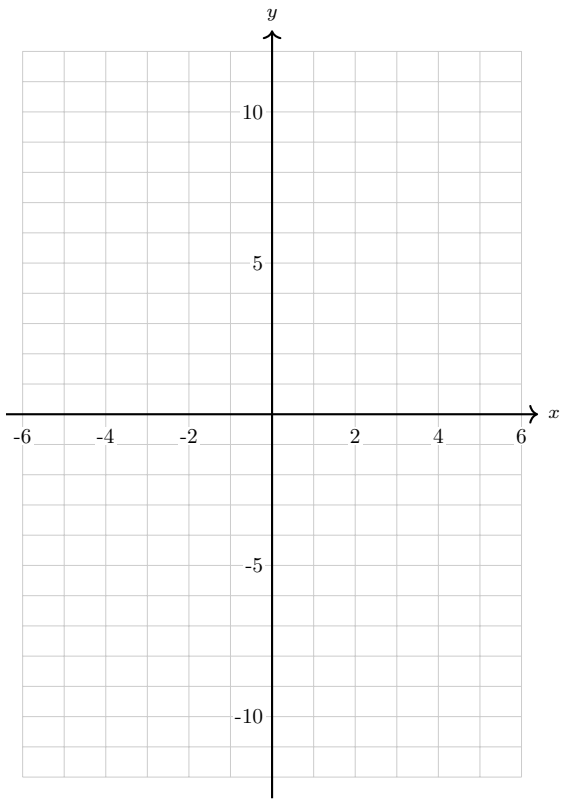
4. From an Equation to a Table and a Graph

Example 3 — A relation whose graph is a straight line

Consider the relation described by the equation $y = 3x - 4$.

(a) Complete the table of values. **(b)** Plot the ordered pairs from (a) on the grid, then connect the points and extend the line with arrows at both ends.

Input (x)	**Output** (y)	**Pair** (x, y)
-2		
-1		
0		
1		
2		



(c) Use the graph to determine the value of y when $x = 5$.

(d) Use the equation to determine the value of y when $x = 5$, and verify your answer in (c).

(e) Write the value of y when $x = 5$ in the table of values, using the first blank row in (a).

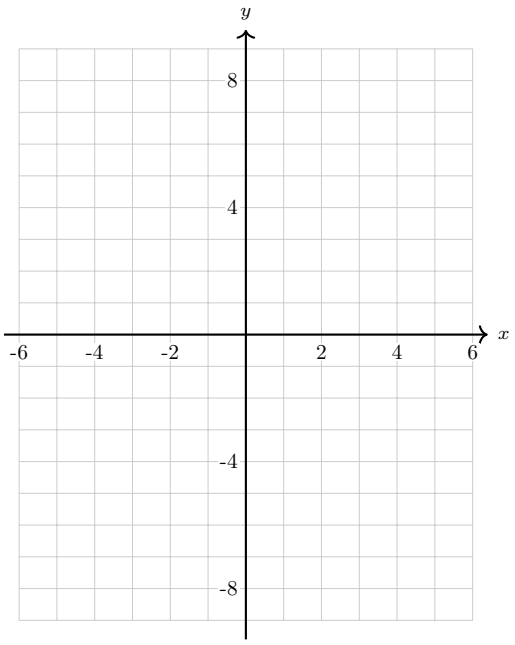
(f) Use the graph to determine the value of x when $y = 8$. Record this ordered pair in the last blank row in (a).

(g) Complete the statement: this relation is called a _____ relation because its graph is a straight line.

Example 4 — A relation whose graph is a curve

Consider the relation described by the equation $y = x^2 - 8$.
(a) Complete the table of values. **(b)** Plot the ordered pairs from (a) on the grid.

Input (x)	**Output** (y)	**Pair** (x, y)
4		
3		
2		
1		
0		
-1		
-2		
-3		



(c) Use the symmetry of the graph or table to predict the value of y when $x = -4$.

(d) Use the equation to determine the value of y when $x = -4$, and verify your answer in (c).

(e) Write the value of y when $x = -4$ in the blank row in (a), then connect the points on the grid with a smooth curve.

(f) Why do you think this type of relation is called a **nonlinear relation**?

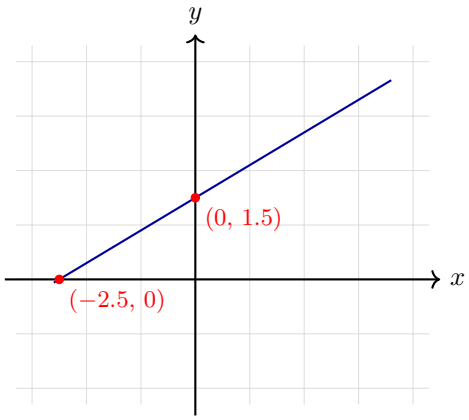
The table test for linearity. When the inputs in a table go up in *equal steps*, look at how the outputs change from one row to the next. If the outputs change by the **same amount** every step (a constant difference), the relation is **linear**. If they do not, it is **nonlinear**. In Example 3 ($x = -2, -1, 0, 1, 2$) the outputs $-10, -7, -4, -1, 2$ go up by 3 every step: linear. In Example 4 ($x = 0, 1, 2, 3, 4$) the outputs $-8, -7, -4, 1, 8$ go up by $1, 3, 5, 7$: nonlinear.

Lesson 3: Where Graphs Meet the Axes

x- and y-intercepts: substitute $y = 0$, substitute $x = 0$ — and read the meaning in context.

1. Inputs, Outputs, and Graphs

Every ordered pair (x, y) of a relation has a first coordinate, the **input** — a value of the **independent variable**, usually x — and a second coordinate, the **output** — a value of the **dependent variable**, usually y, because its value *depends on* the input chosen. Graphing all the ordered pairs traces out a curve (a line, a parabola, a circle), and two places on that curve are worth special attention: where it crosses the horizontal axis, and where it crosses the vertical axis.



The line above crosses the x-axis at $(-2.5, 0)$ and the y-axis at $(0, 1.5)$. These points are called **intercept points**.

2. Definitions and the Algebraic Method

x-intercept: the x-coordinate of a point where a graph **meets** the x-axis — it may cross the axis or just touch it (the parabola $y = x^2$ touches the x-axis at the origin). At every such point the y-coordinate equals zero.

y-intercept: the y-coordinate of a point where a graph meets (crosses or touches) the y-axis. At every such point the x-coordinate equals zero.

Value or point? In this unit the **intercept** is the *value* and the **intercept point** is the *ordered pair*: for the line above, the x-intercept is -2.5 and the x-intercept point is $(-2.5, 0)$.

To determine intercepts algebraically, given the equation of a relation: to find the x-intercept(s), substitute $y = 0$ and solve for x; to find the y-intercept(s), substitute $x = 0$ and solve for y. The equation of a relation can be written in several equivalent forms — $y = \frac{4}{3}x + 8$ and $4x - 3y + 24 = 0$ describe the very same line — and whatever the form, the same substitution strategy finds the intercepts.

Example 1 — A line with both variables on one side

A line has equation $4x - 3y + 24 = 0$. Algebraically determine the x-intercept and the y-intercept of the line. Write each intercept point as an ordered pair.

x-intercept **y-intercept**

Example 2 — A circle has two of each

The equation of a circle is $x^2 + y^2 = 36$. Calculate the x-intercept(s) and y-intercept(s) of its graph. Write each intercept point as an ordered pair.

x-intercept(s) **y-intercept(s)**

Example 3 — A parabola: factor to find the x-intercepts

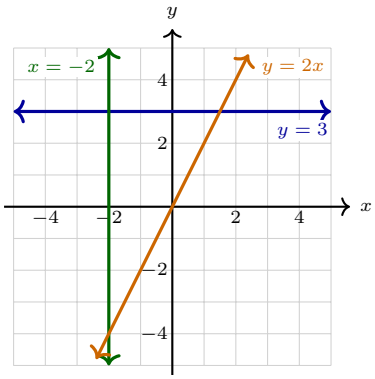
A parabola has equation $y = x^2 - 2x - 24$. Algebraically determine the x-intercept(s) and the y-intercept of the parabola. Write each intercept point as an ordered pair.

x-intercept(s) **y-intercept**

3. Lines With One, Two, or Infinitely Many Intercepts

Most lines cross both axes, so they have two intercept points. Some lines do not.

Example 4 — Horizontal, vertical, and through-the-origin lines



For each line, state the x-intercept(s) and the y-intercept(s), if any, and the number of intercept points. (Line (c), $y = 0$, is the x-axis itself.)

(a) $y = 3$

(b) $x = -2$

(c) $y = 0$

(d) $y = 2x$

Summary. A line can have **two** intercept points; **exactly one** (a horizontal or vertical line that is not an axis, or a slanted line through the origin); or **infinitely many** (the line is the x-axis, $y = 0$, or the y-axis, $x = 0$).

4. Interpreting Intercepts in Applied Situations

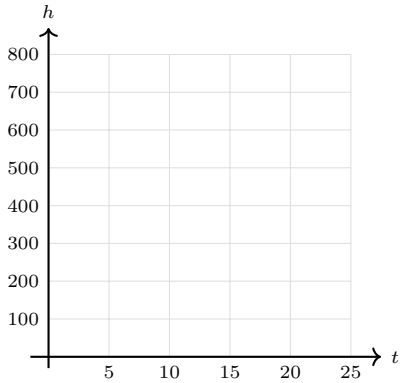
When a relation models a real situation its intercepts carry meaning tied to the context. Reading a value off a graph is **interpolation** when the value lies between two plotted points and **extrapolation** when the graph must be extended beyond them. Substituting into the formula always gives an **exact** value; reading from a graph gives only an **estimate**.

Example 5 — A descending hot-air balloon

A hot-air balloon begins a steady descent. Its height above the ground, h metres, after t minutes ($t \geq 0$) is given by $h = 800 - 40t$.

(a) Complete the table of values, then plot the ordered pairs on the grid and connect them with a straight line.

Input (t)	Output (h)	(t, h)
0		
5		
10		
15		



(b) What does the ordered pair $(0, 800)$ represent in the context of this problem?

(c) Determine the t-intercept of the graph algebraically. What does the t-intercept represent in the context of this problem?

(d) Use the formula to calculate the exact height of the balloon after 12 minutes.

(e) Use the formula to determine how many minutes it takes the balloon to reach a height of 160 metres.

(f) The table stops at $t = 15$. Is finding the height at $t = 12$ in (d) an example of interpolation or extrapolation? What about finding the time in (e)?

Lesson 4: Determining the Domain and Range of a Relation

The set of inputs and the set of outputs — read from ordered pairs, tables, arrow diagrams, graphs and real situations.

1. Domain and Range Defined

The **domain** of a relation is the set of all values taken by the independent variable — the inputs, usually x. The **range** is the set of all values the relation actually produces for the dependent variable — the outputs, usually y. A relation may be presented as a set of ordered pairs, a table of values, an arrow diagram (mapping diagram), an equation, a graph, or in words; whichever form is given, the same two questions apply: *which input values are used?* and *which output values are produced?*

How the answer is written. A relation with finitely many values gets a **list** inside braces, each value written once and in increasing order, e.g. $D = \{2, 5, 9\}$ (a finite set too long to write out may use set-builder notation with the right number set instead, as in Example 5). A relation with infinitely many values gets **set-builder notation**, e.g.

$$R = \{y \mid y \geq -3,\ y \in \mathbb{R}\},$$

read "the set of all y such that $y \geq -3$ and y is a real number." The symbols N, W, I, Q and $\mathbb{R}$ name the number set the variable is drawn from. In this unit the real numbers are written $\mathbb{R}$ (double-struck), so they are never confused with R for a range. For a **count**, use W (whole numbers) — or I together with a lower bound such as $n \geq 0$ — because I on its own includes negative numbers.

Example 1 — Domain and range from ordered pairs

List the domain and range of each set of ordered pairs.

(a) $(2, 3),\ (0, 7),\ (4, 10),\ (6, 11),\ (-2, 3)$

(b) $(5, 5),\ (0, 5),\ (-5, 5),\ (3, 11),\ (-9, 5)$

2. Reading Domain and Range from a Graph

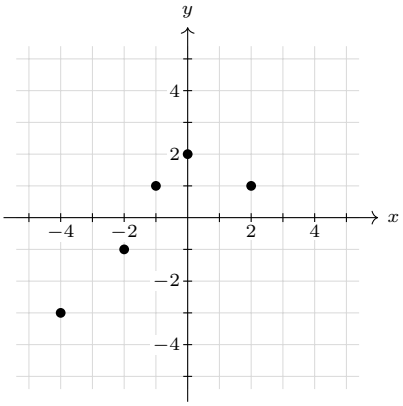
The domain is the shadow the graph casts on the x-axis when a light shines straight down onto it; the range is the shadow it casts on the y-axis when a light shines straight across (both shadows are shaded on Example 2(b) below). Three features of the drawing decide the answer.

- **Isolated (discrete) points** give a list: the x-coordinates plotted are the domain, the y-coordinates plotted are the range, each value written once.
- **An arrow** on an end means the graph continues forever in that direction, so that side is unbounded: an unbounded domain is written $\{x \mid x \in \mathbb{R}\}$, and a range bounded on one side only is written $\{y \mid y \leq 3,\ y \in \mathbb{R}\}$ or $\{y \mid y \geq -3,\ y \in \mathbb{R}\}$.
- **A solid endpoint** belongs to the graph, so its value is included and the inequality uses $\leq$ or $\geq$; a **hollow endpoint** does not belong to the graph, so its value is excluded and the inequality uses $<$ or $>$.

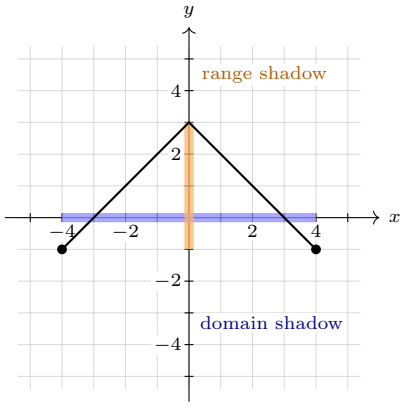
Example 2 — Domain and range from a graph

State the domain and range of each relation graphed below.

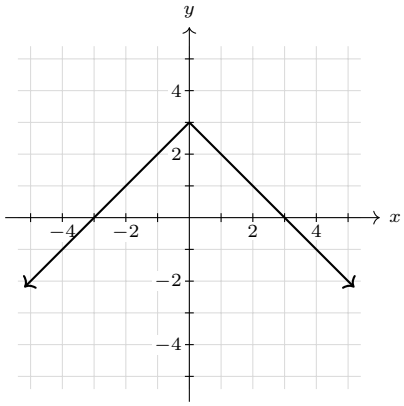
(a)



(b)



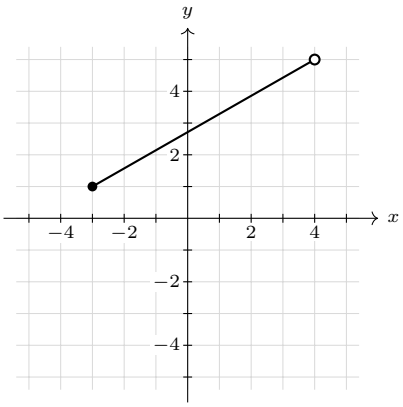
(c)



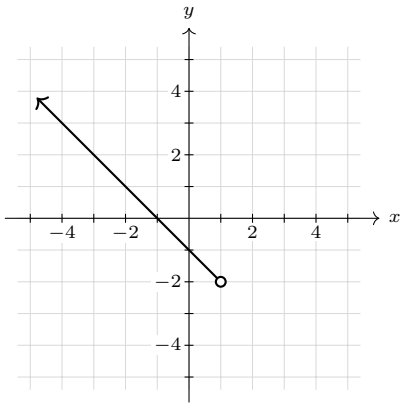
Example 3 — When an endpoint is open

State the domain and range of each relation. Watch for the **hollow** endpoints.

(a) A segment from a solid endpoint at $(-3, 1)$ to a hollow endpoint at $(4, 5)$.



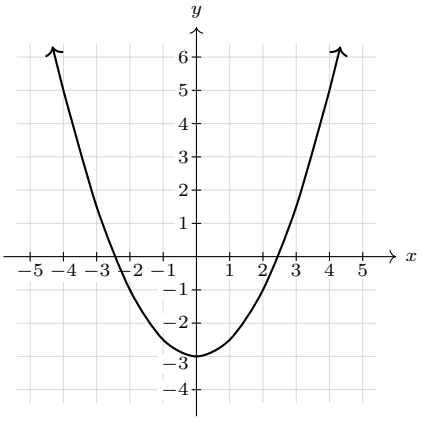
(b) A ray that starts at a hollow endpoint at $(1, -2)$ and continues up to the left forever.



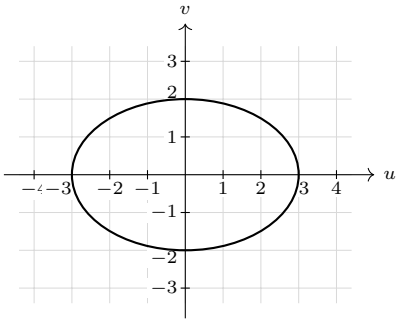
Example 4 — Curves, ellipses, arrow diagrams and a circle

State the domain and range of each relation described below.

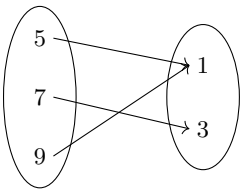
(a)



(b) An ellipse, with horizontal axis u and vertical axis v.



(c)



(d) A circle with centre $(3, -2)$ and a radius of 6.

3. Domain and Range in Real-World Situations

When a relation models a real situation, its domain and range are cut down to the values that make physical sense: time cannot be negative, and a count of people must be a whole number, so its domain is written with $\in W$ rather than $\in \mathbb{R}$. These practical restrictions make the domain and range of an applied relation narrower than those of the same equation treated as pure mathematics.

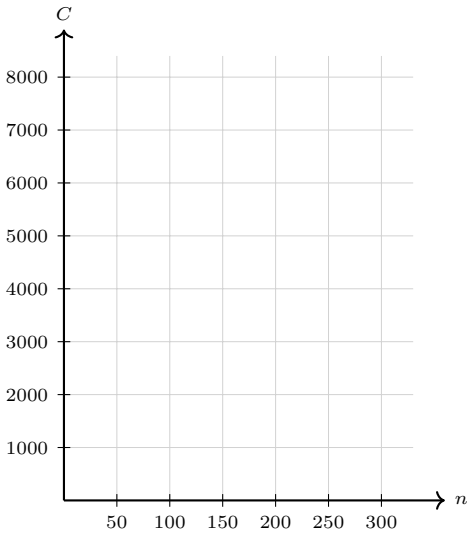
Example 5 — A rental cost with a practical domain

A community league is renting a hall for a fundraiser dinner. The venue charges a setup fee of \$350 plus \$25 per guest, so $C = 350 + 25n$ gives the total rental cost, in dollars, for n guests. The venue requires at least 50 guests and holds at most 300.

(a) Complete the table of values for $n = 50,\ 100,\ 150,\ 200,\ 250$ and 300 guests.

n (guests)	50	100	150	200	250	300
C (dollars)						

(b) Plot the six ordered pairs from part (a) on the grid.



(c) If all possible ordered pairs from part (b) were plotted, state the domain and range of the relation, and explain why there are restrictions on both.

Lesson 5: What Is a Function?

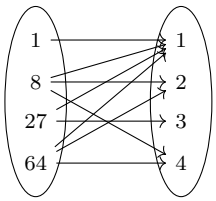
The function test in every representation, and the vertical line test.

1. What Is a Function?

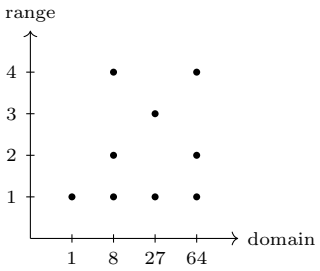
Function. A **function** is a relation in which each element of the domain is paired with exactly one element of the range. If any element of the domain is paired with more than one element of the range, the relation is **not** a function.

Take the domain $D = \{1, 8, 27, 64\}$ and the range $R = \{1, 2, 3, 4\}$, joined by two relations. For relation (i) we ask, for each domain element, which range elements divide it evenly; for relation (ii) we ask which range element, when cubed, produces the domain element.

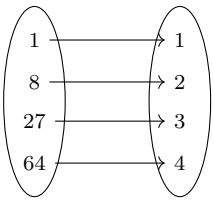
Relation (i): "is a multiple of"



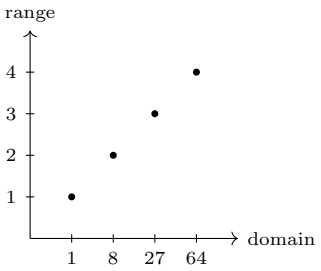
$(1,1),\ (8,1),\ (8,2),\ (8,4),\ (27,1),\ (27,3),\ (64,1),\ (64,2),\ (64,4)$



Relation (ii): "is the cube of"

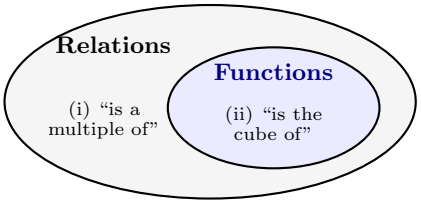


$(1,1),\ (8,2),\ (27,3),\ (64,4)$



In relation (i) the domain element 8 is paired with three different range elements (1, 2 and 4), so relation (i) is **not** a function. In relation (ii) every domain element is paired with exactly one range element, so relation (ii) **is** a function.

Functions are relations. Every function is a relation, but **not every relation is a function**. A function is a special kind of relation — one in which no input has more than one output. In the diagram, relation (ii) belongs inside the inner oval; relation (i) is a relation but sits outside the "Functions" oval.



Example 1 — Reading the function test off three representations

Using the two relations above, explain how you can tell whether a relation is a function by examining each of the following.
(a) the arrow diagram **(b)** the list of ordered pairs **(c)** the graph

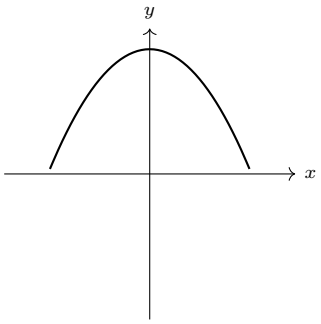
2. The Vertical Line Test

Vertical line test. If every vertical line crosses the graph at most once, the relation **is** a function. If any vertical line crosses the graph more than once, the relation is **not** a function. (A vertical line outside the domain misses the graph completely — that is allowed.)

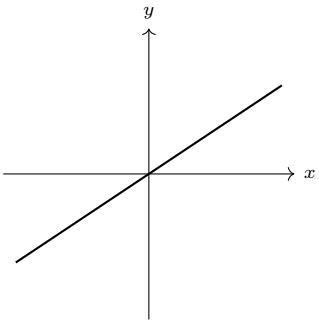
Example 2 — Four relations, one test

Each of the following is the graph of a relation. **(a)** Classify each statement in the table below as true (T) or false (F) for each graph.

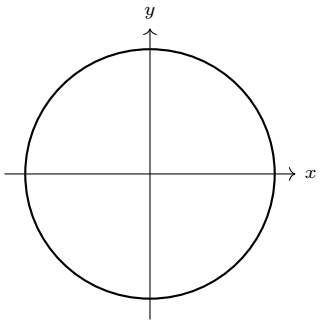
Graph A



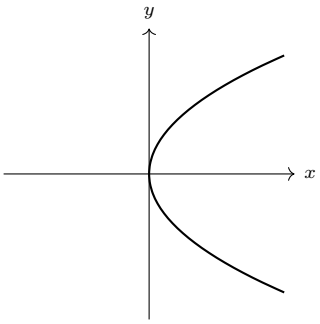
Graph B



Graph C



Graph D



	A	**B**	**C**	**D**
For each input value there is only one output				
For each output value there is only one input				
The relation is a function				

(b) From graph C, write two ordered pairs which show that the relation is not a function.

(c) From graph D, write two ordered pairs which show that the relation is not a function.

(d) On graphs A and B, imagine a series of vertical lines. Do any of these lines intersect the graph of the relation at more than one point? Explain what this tells you about A and B.

3. Recognizing Functions in Different Forms

A relation may be given as a list of ordered pairs, a table, an arrow diagram (mapping diagram), a graph or a description in words. The test is the same in every case: does every element of the domain appear with exactly one output?

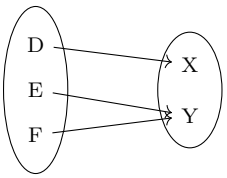
Example 3 — Six relations in six forms

Determine which of the following relations are functions. Justify each answer.

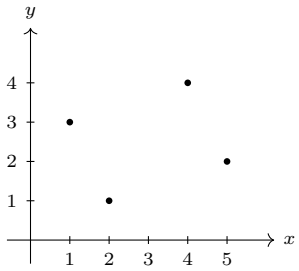
(a) $(7,3),\ (5,7),\ (-2,9),\ (0,3),\ (5,4)$

(b) $(4,4),\ (1,3),\ (6,9),\ (-5,4)$

(c)

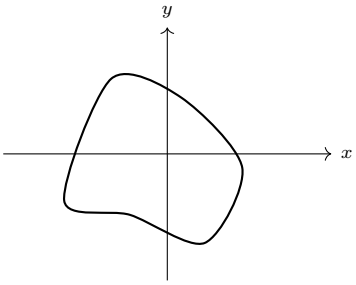


(d)



(e) The relation pairing each vehicle registered in Alberta with its licence plate number.

(f)



Lesson 6: Function Notation

Naming a function, writing $f(x)$, evaluating $f(a)$ and solving $f(x) = b$. (Extension: mapping notation; substituting expressions.)

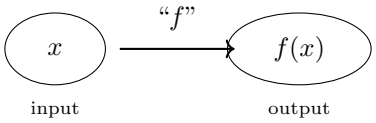
In Lesson 5 you decided whether a relation is a function. Once we know a relation is a function, we can give it a name, such as f, and a short way to write "the output when the input is x": $f(x)$. This lesson is about reading, writing and using that notation.

1. Function Notation

For the function $f(x) = 3x + 1$, the output when the input is 4 is

$$f(4) = 3(4) + 1 = 13.$$

We say the **image** of 4 under f is 13, or that the **value of the function** is 13 when $x = 4$. The symbol $f(x)$ is read "f of x": an input x enters the function machine f and the output is labelled $f(x)$.



The two columns below show the same information written two ways.

function notation	equation of the graph
$f(x) = 3x + 1$	$y = 3x + 1$
$f(4) = 3(4) + 1$	$y = 3(4) + 1$
$f(4) = 13$	$y = 13$ (the point $(4, 13)$ is on the graph)

The symbol $f(x)$ therefore does double duty: it is a *formula* for the function f, and it also represents the *output value* of the function for a particular input x.

Key facts about function notation.

- $f(x)$ does not mean "f times x" — the parentheses indicate an input, not multiplication.

- The letter in front of the parentheses (here, f) is simply the **name** of the function; other common names are g, h, P, and so on.
- Inputs (values of the independent variable, usually x) are plotted on the **horizontal** axis.
- Outputs (values of the dependent variable, usually y or $f(x)$) are plotted on the **vertical** axis.

2. One Function, Several Representations

Example 1 — Completing a function and showing it three ways

(a) The relation "is the cube of" from Lesson 5 is a function. Written in function notation it is $g(x) = \sqrt[3]{x}$, with domain $\{1, 8, 27, 64\}$. Complete:

$$g(1) = _____ \qquad g(8) = _____ \qquad g(27) = _____ \qquad g(64) = _____$$

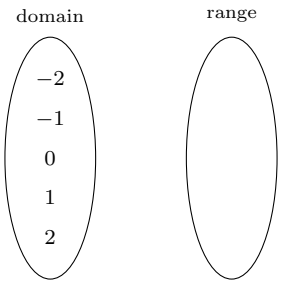
(b) Consider the function $f(x) = 3x - 2$, with domain $\{-2, -1, 0, 1, 2\}$.
(i) Complete:

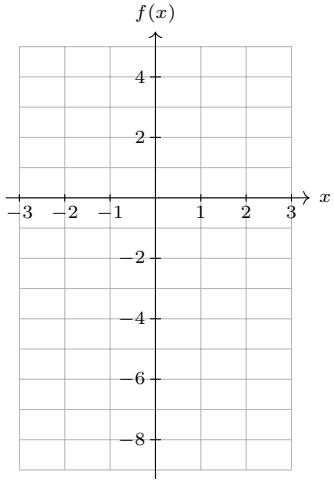
$$f(-2) = _____ \qquad f(-1) = _____ \qquad f(0) = _____ \qquad f(1) = _____ \qquad f(2) = _____$$

(ii) List the elements of the range of the function.

(iii) Show the function f as an arrow diagram, a set of ordered pairs, and a graph.

Ordered pairs:





3. Evaluating and Solving

To **evaluate** $f(a)$ for a number a, replace every x in the formula for $f(x)$ with a, then simplify. For a **linear** function the input can also be an expression: if $g(x) = 5x - 2$, then $g(3t) = 5(3t) - 2 = 15t - 2$. To **solve** $f(x) = b$, set the formula equal to b and solve for x — you are given the output and asked for the input.

Example 2 — Two functions, five values

Consider the functions $f(x) = x^2 + 3$ and $g(x) = 5 - x$. Evaluate each of the following.

(a) $f(4)$ **(b)** $g(2)$ **(c)** $f(-3)$

(d) $g(-1)$ **(e)** $f(0) - g(0)$

Example 3 — The same function asked for in three ways

Consider the function f defined by $f(x) = 4x^3 - 3x$, $x \in \mathbb{R}$. Determine each of the following.

(a) $f(-2)$ **(b)** the value of f when $x = 3$ **(c)** the image of 5 under f

Example 4 — Substituting into a linear function

Consider the linear function $g(x) = 5x - 2$.

(a) an expression for $g(a)$ **(b)** an expression for $g(3t)$ **(c)** an expression for $g(a + 1)$ **(d)** Solve $g(x) = 18$.

Example 5 — Going backwards: solving $f(x) = k$

Consider the function $f(x) = x^2 - 7$, $x \in \mathbb{R}$.

(a) Evaluate $f(5)$. **(b)** Solve the equation $f(x) = 29$. **(c)** Solve $f(t) = 38$, where $t > 0$. Give your answer in simplest radical form.

4. Extension — Mapping Notation and Substituting Expressions

This section goes beyond the Math 10C outcomes.

Mapping notation. Some books write a function as a *mapping*: $f : x \to 3x + 1$ ("f maps x to $3x + 1$"), so $f : 4 \to 13$ means the same as $f(4) = 13$.

mapping notation	function notation	equation of the graph
$f : x \to 3x + 1$	$f(x) = 3x + 1$	$y = 3x + 1$
$f : 4 \to 13$	$f(4) = 13$	$y = 13$

Substituting an expression into a non-linear function. To find $f(3x)$ or $f(x + 2)$, replace every x in the formula with the *entire* expression, in parentheses, then simplify.

Example 6 — *(Extension)* Expressions in a cubic function

For $f(x) = 4x^3 - 3x$ from Example 3, determine:

(a) an expression for $f(a)$ **(b)** an expression for $f(3x)$

Example 7 — *(Extension)* Substituting a binomial

If $P(x) = 3x^2 - 5x + 2$, determine a simplified expression for $P(x + 2)$.

Lesson 7: Function Notation: Graphs, Conversions, and Real-World Models

Converting between $y = \ldots$ and $f(x) = \ldots$, reading $f(a)$ and solving $f(x) = b$ off a graph, and interpreting a model in context.

1. Function Notation and x-y Notation

Every function written with y can also be written in function notation, and the reverse: the two forms describe the same rule. In $f(x) = 2x - 3$ the input values (values of x) form the domain and the output values (values of $f(x)$) form the range, so $f(x)$ plays exactly the role y plays.

Function notation	x-y **notation**
$f(x) = 2x - 3$	$y = 2x - 3$

To go from function notation to x-y notation, replace the function name and its parentheses — such as $f(x)$ — with y. To go the other way, replace y with a function name and parentheses. The independent variable never changes: $h(t) = t^2 + 3t - 8$ becomes $y = t^2 + 3t - 8$.

- Values of the **independent variable** (the input) make up the **domain**, and are shown on the **horizontal axis**.
- Values of the **dependent variable** (the output) make up the **range**, and are shown on the **vertical axis**.

Example 1 — Converting between the two forms

(a) Express each relation given in function notation as an equation in two variables.

i) $f(x) = 5x - 17$

ii) $h(t) = t^2 + 3t - 8$

(b) Express the relation $y = 9x - 4$ in function notation.

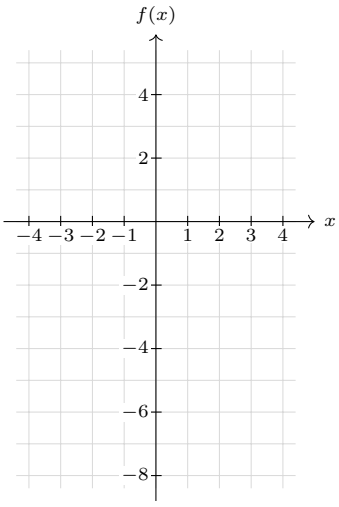
(c) The graph of the function defined by $y = f(x)$ has equation $y = 7 - 2x$. Express the equation in function notation.

2. Graphing a Function

Once a function is written in x-y notation it is graphed like any other equation in two variables: choose input values, compute the matching output values, plot the ordered pairs, and join them.

Example 2 — From a table of values to a graph

Use a table of values to sketch the graph of the function $f(x) = 2x - 3$.



3. Reading Values from the Graph of a Function

Every question about inputs, outputs, ordered pairs, domain and range can be answered from the picture. Two opposite readings are needed, and they must not be confused.

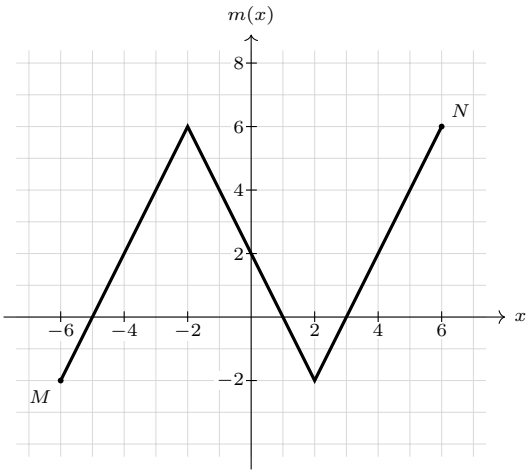
- $f(3)$ gives the input $x = 3$ and asks for the output: move along the horizontal axis to 3, then read *up or down* to the graph. There is exactly one answer.

- $f(x) = 3$ gives the output 3 and asks for the input: move along the vertical axis to 3, then read *across* to the graph. There may be one answer, several, or none at all — count every point of the graph at that height.

The domain is the span of the graph along the horizontal axis; the range is its span along the vertical axis. An x-intercept is a point where the output is 0, so it is written $f(a) = 0$; the y-intercept is the output at $x = 0$, written $f(0)$.

Example 3 — Reading a graph in both directions

The graph of a function m is shown below.



(a) Complete:
i) $m(-3) =$ _____
ii) $m(1) =$ _____
iii) $m(4) =$ _____

(b) Write the ordered pairs associated with i), ii) and iii) above.

(c) State the value(s) of x if **i)** $m(x) = 6$, **ii)** $m(x) = 0$, **iii)** $m(x) = 8$.

(d) Use the notation in (a) to make a statement about the points M and N on the graph.

(e) Write the x- and y-intercepts of the graph using function notation.

(f) Complete the following statements.
The domain of m is
$\{x \mid ____ \leq x \leq ____,\ x \in \mathbb{R}\}$.
The range of m is
$\{m(x) \mid ____ \leq m(x) \leq ____,\ m(x) \in \mathbb{R}\}$.

4. Replacing an Equation with Function Notation

A scenario described by an equation such as $h = -2t + 12$, where h is the height of a candle, in cm, after t hours, is a function: every value of t produces exactly one value of h. Renaming the output variable to show its dependence on the input gives

$$h(t) = -2t + 12.$$

Nothing about the rule has changed — only how it is written. The expression $h(t)$ is read "h of t" and means the output produced by h when the input is t, so $h(4)$ is a compact way of asking for the height of the candle after 4 hours.

Reading function notation. In a statement $f(a) = b$, a is the **input** (independent variable) and b is the **output** (dependent variable) that results from it. To *evaluate* $f(a)$, substitute a for the input variable everywhere it appears in the rule, then simplify. To *interpret* $f(a) = b$ in words, describe both the input and the output in the context of the problem — units and all.

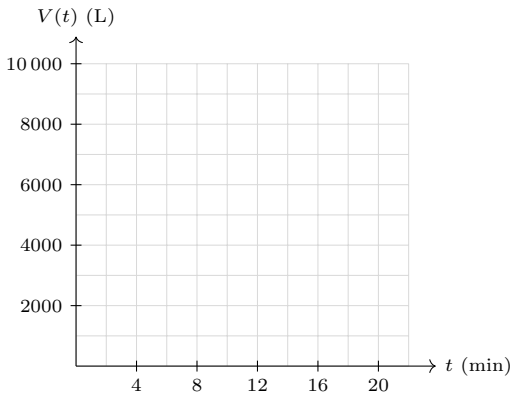
5. Domain, Range, and the Limits of a Model

A rule such as $h(t) = -2t + 12$ is defined for every real number t, but in a real-world scenario only some inputs make sense: a candle cannot have a negative height and time cannot run backward. The domain and range of the real-world function are therefore usually restricted to a smaller interval than the bare rule allows. Identifying where a model stops describing the physical situation — for instance once the candle has completely melted — matters as much as evaluating it correctly.

Example 4 — Draining a pool

A community pool is being drained for the winter. The volume of water remaining is modelled by $V(t) = -450t + 9000$, where $V(t)$ is the volume of water, in litres, remaining in the pool t minutes after draining begins.

(a) Sketch the graph of the function on the grid provided.



(b) Determine the value of $V(6)$.

(c) Write in words the meaning of $V(6)$.

(d) Evaluate the following, and explain the significance of each.

i) $V(0)$ **ii)** $V(15)$ **iii)** $V(22)$

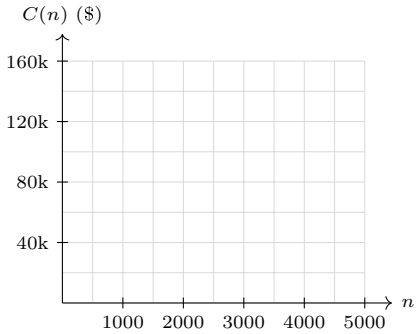
(e) How long will it take for the volume to drop to 1350 litres?

(f) Suggest an appropriate domain and range for the function.

Example 5 — Cost of production

Bright Spark Electronics manufactures wireless earbuds. The cost of production is modelled by $C(n) = 18\,400 + 27n$, where $C(n)$ is the cost, in dollars, of producing n pairs of earbuds.

(a) Sketch the graph of the function for a maximum of 5000 pairs.



(b) Determine the value of $C(120)$.

(c) Write in words the meaning of $C(120)$.

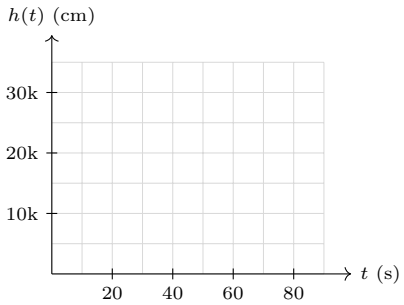
(d) Evaluate $C(0)$ and explain its significance.

(e) How many pairs of earbuds can be produced for \$29 200?

Example 6 — A model rocket

A model rocket is launched straight up from the ground. Its height is modelled by $h(t) = -20t^2 + 1600t$, where $h(t)$ is the height, in cm, of the rocket t seconds after launch.

(a) Sketch the graph of the function on the grid provided.



(b) Determine the values of $h(20)$ and $h(60)$.

(c) Does $h(20) = h(60)$? Do they mean the same thing? Explain.

(d) Evaluate the following, and explain their significance in the context of the question.

i) $h(0)$ **ii)** $h(80)$ **iii)** $h(90)$

(e) What is the highest point the rocket will reach?

(f) When will the rocket land?

(g) Suggest an appropriate domain and range for $h(t)$.

Lesson 8: Interpreting and Sketching Real-World Graphs

Reading a graph's story — increasing, decreasing, constant, and rate of change — then sketching a shape from a written description.

1. Increasing, Decreasing, and Constant Behaviour

A real-world graph plots a changing quantity (vertical axis) against time or another variable (horizontal axis). Every such graph breaks into intervals where the quantity is **increasing** (the graph rises left to right), **decreasing** (it falls), or **constant** (it is flat).

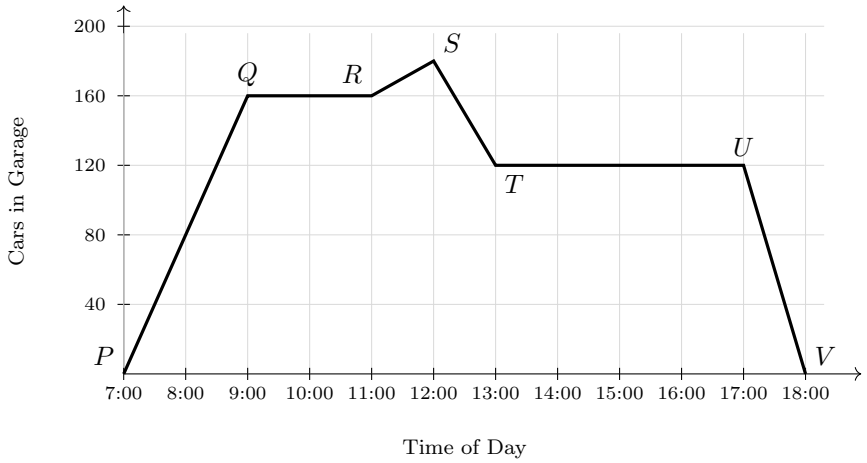
The **steepness** of a segment measures how quickly the quantity changes — its **rate of change**:

$$\text{rate of change} = \frac{\text{change in the quantity (vertical)}}{\text{change in time (horizontal)}}.$$

A larger magnitude of steepness, positive or negative, means a *faster* rate of change; a flat segment has a rate of change of zero.

Example 1 — Reading the story of a parking-garage graph

A downtown parking garage opens empty at 7:00 and closes for the night at 18:00. The graph shows the number of cars parked in the garage through the day. It is made of six line segments meeting at P, Q, R, S, T, U, V.



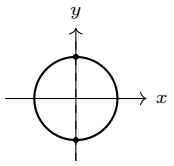
(a) With reference to the parking garage, explain what is happening during each interval.
(i) P to Q **(ii)** R to S **(iii)** T to U

(b) Determine the rate of change, in cars per hour, between S and T.

(c) Which single segment shows the fastest rate of change, and what does this tell you about the garage at that time?

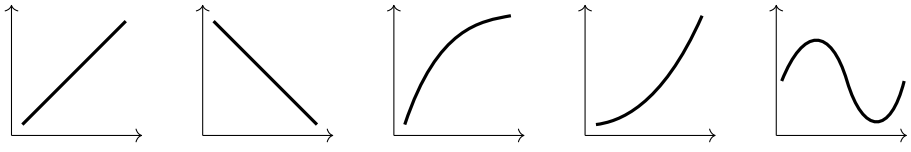
(d) For how many hours in total was the number of cars in the garage increasing?

The vertical line test, revisited. A graph represents a quantity as a **function** of time only if every vertical line crosses it at most once; otherwise a single instant would give more than one value of the quantity, which is impossible. The circle at right fails the test — a vertical line through its centre meets the curve twice — so it cannot show y as a function of x. Every graph in this lesson that plots a quantity against *time* must pass this test: an object can only be in one state at any one instant.



2. Matching a Description to the Shape of a Graph

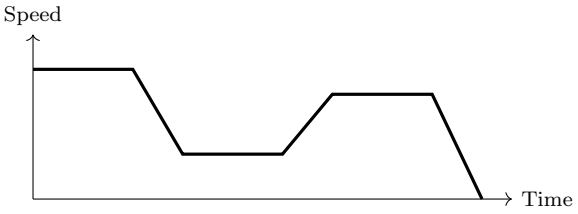
Before sketching from words it helps to recognize a small vocabulary of common shapes.



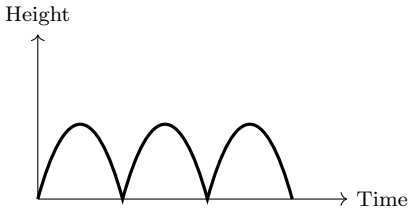
Example 2 — From a shape back to a story

Suggest a possible real-world scenario for each graph below.

(a)



(b)



3. Sketching a Graph From a Description

To sketch a graph with no numerical scale, focus only on its overall *shape*: where it rises, falls, stays flat, or repeats, and how sharply it does so. Exact numbers do not matter — getting the pattern of behaviour right does.

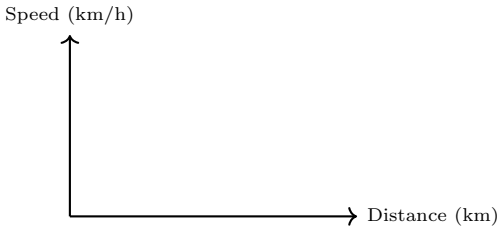
Example 3 — From a story to a shape

Sketch a graph with no scale for each situation.

(a) The temperature inside a kiln while pottery is fired: the kiln heats up steadily, holds at a high temperature for a long soak, then is turned off and cools back toward room temperature.



(b) The speed of a cyclist on a flat trail with one steep hill in the middle: she rides at a steady speed, slows sharply climbing the hill, nearly stops at the crest, then speeds up quickly down the far side before levelling off.



Lesson 9: Relations and Functions Unit Review

Cumulative review — discrete and continuous variables, domain and range, intercepts, function notation, and the vertical line test.

1. Variables, Domain, and Range

A variable is **discrete** if it can take only separate, countable values, such as a number of items; it is **continuous** if it can take any value within an interval, such as a measurement of time, length or volume. The **domain** of a relation is the set of values taken by the independent variable — the inputs, the first coordinates, plotted on the horizontal axis. The **range** is the set of values taken by the dependent variable — the outputs, the second coordinates, plotted on the vertical axis.

How the answer is written. A relation with finitely many values gets a **list** inside braces, each value written once and in increasing order, e.g. $D = \{2, 5, 9\}$. A relation with infinitely many values gets **set-builder notation**, e.g. $R = \{y \mid y \geq -3,\ y \in \mathbb{R}\}$ (here R names the range and $\mathbb{R}$ the real numbers); a set unbounded in both directions is written $\{x \mid x \in \mathbb{R}\}$. For a count, use W (whole numbers) — or I with a lower bound such as $n \geq 0$ — in place of $\mathbb{R}$.

Example 1 — Discrete or continuous

Classify each variable as discrete or continuous.

(a) The number of students in a classroom.

(b) The time taken to swim 100 metres.

(c) The volume of water in a tank.

(d) The number of pages in a book.

Example 2 — Finding the false statement

Exactly one of the following statements is **false**. Identify it and explain why.

(a) The domain of a relation is the set of values taken by the independent variable.

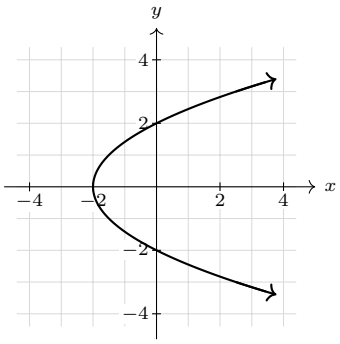
(b) The independent variable is represented by the second coordinate of each ordered pair.

(c) On a Cartesian plane, the outputs of a relation are graphed on the vertical axis.

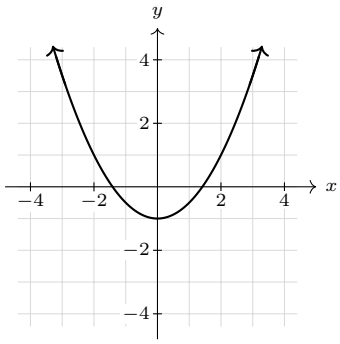
(d) The range of a relation is the set of values taken by the dependent variable.

Example 3 — Domain and range from a graph

The graphs of two relations, P and Q, are shown below. Both curves continue forever in the directions of the arrows. State the domain and range of each relation, then say whether the two domains are the same or different, and whether the two ranges are the same or different.



Relation P



Relation Q

2. Intercepts of a Relation

An **x-intercept** is the value of x where the graph meets (crosses or touches) the x-axis: substitute $y = 0$ and solve for x. A **y-intercept** is the value of y where the graph meets the y-axis: substitute $x = 0$ and solve for y. As in Lesson 3, the *intercept* is the value and the *intercept point* is the ordered pair. A relation may have several x-intercepts, but a function has at most one y-intercept, since one input cannot give two outputs.

Example 4 — An intercept to the nearest hundredth

To the nearest hundredth, determine the x-intercept of the graph of the line $3x + 7y - 10 = 0$.

Example 5 — Combining both kinds of intercept

The graph of the relation $5x^2 - 4y = 20$ has x-intercepts a and b, and y-intercept c. Determine the value of the product abc.

3. Functions in Context

A relation is a **function** only when every input produces exactly one output. On a graph this is checked with the **vertical line test**: if every vertical line crosses the graph at most once, the relation is a function; if any vertical line crosses it more than once, it is *not*. Function notation $f(x)$, read "f of x," names the output that the function f produces from the input x; it does not mean f times x. To evaluate $f(a)$, replace every x in the rule for f with a and simplify.

Example 6 — A cost model, forwards and as a domain and range

The cost, in dollars, of publishing a school newsletter is partly fixed and partly depends on the number n of copies printed: $C = 150 + 0.08n$.

(a) Determine the cost of publishing 685 newsletters.

(b) The domain of this relation can be written in the form $n \geq a$ and the range in the form $C \geq b$. State the values of a and b.

Example 7 — Evaluating over a finite domain

The function $f(x) = 2x^2 - 5$ has domain $\{-5, -3, 0, 1, 4\}$. Evaluate f at each value.

(a) $f(-5) =$

(b) $f(-3) =$

(c) $f(0) =$

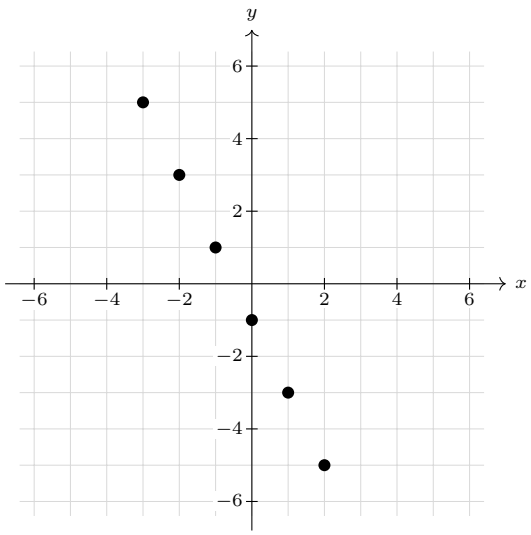
(d) $f(1) =$

(e) $f(4) =$

(f) State the range of the function.

Example 8 — Reading function values off a graph

The diagram shows the graph of the function $y = f(x)$. The function is defined only at the six inputs shown, so the graph is six separate dots, not joined. Use the graph to state the value of $f(-2) + f(2)$.



4. Working with Linear Functions

A linear function can be used **forwards**, by substituting a given input to get the output, or **backwards**, by setting the function equal to a given output and solving the resulting equation for the input. The graph of a linear function such as $Q(x) = 6x + 5$ is a straight (non-vertical) line, so it always passes the vertical line test.

Example 9 — Forwards and backwards through a linear function

Consider the function $Q(x) = 6x + 5$.

(a) Determine the value of $Q(4)$.

(b) If $Q(b) = 2$, determine the value of b.

Example 10 — The vertical line test

Which of the diagrams below shows a relation that is **not** a function? Justify your answer using the vertical line test.

